
Ion-driven Electron Scattering and Resistance Modulation in TiO₂ Memristors

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Abstract

This study proposes a physical model for TiO₂-based memristors that links ionic motion to microscopic electron transport. Unlike conventional models that treat resistance as a simple geometric state, we model oxygen vacancies as charged ions (+2*q*) and analyze their role as dynamic scattering centers. By integrating a thermally activated ion hopping model with Drude transport theory, we demonstrate that the spatiotemporal redistribution of ions determines the electron scattering time (τ), which in turn modulates the macroscopic resistance. Numerical simulations successfully reproduce the characteristic i-v hysteresis loops and provide a quantitative framework for understanding the interplay between ionic drift and resistive switching in memristive systems.

We reinterpret the memristive state as the average of electronic scattering time. Using 2 phenomenological postulates and the relation introduced in previous studies, the scattering time is shown to be a closed function of charge, consistent with the defining property of memristive systems.

1 Introduction

The general definition of a memristive system is given by the following relations between voltage and current [1]:

$$v = R(w, i)i, \quad \frac{dw}{dt} = f(w, i) \quad (1.1)$$

Here, w denotes an internal state variable that characterizes the state of the system. Considerable efforts have been devoted to realizing such systems physically, among which a representative example is the Strukov model memristive system based on ionic motion in solid-state materials [2].

In the Strukov model, a memristor consisting of a TiO₂ thin film sandwiched between two metal electrodes is proposed. The TiO₂ layer is divided into two regions with distinct electrical properties: a doped region with relatively high conductivity and an undoped region with low conductivity. By modeling each region as a resistor, the entire memristor can be described as a series connection of the resistance of the doped region R_{on} and that of the undoped region R_{off} . Defining the length of the doped region as the state variable w the total resistance w is expressed as

$$R(w) = R_{on} \frac{w}{D} + R_{off} \left(1 - \frac{w}{D}\right) \quad (1.2)$$

where D is the total of the TiO₂ layer.

The doping referred to in this context does not correspond to conventional chemical impurity doping, but rather to defect-induced self-doping in TiO₂. Subsequent studies have interpreted the doped

region as an oxygen-vacancy-rich TiO_{2-x} phase. Oxygen vacancies are generated when TiO_2 crystals are exposed to oxygen-deficient environments at temperatures above approximately 400°C [3]. Since this process is reversible, oxygen vacancies can be metastable at room temperature depending on local chemical environments and defect complexes. To stabilize them, donor-type dopants such as Fe^{3+} can be introduced, enabling the retention of oxygen vacancies even at ambient conditions [4]. First-principles calculations indicate that the most stable charge state of oxygen vacancies in TiO_2 corresponds to an effective charge of $+2q$ [5].

In most previous studies, the temporal evolution of the state variable w has been attributed to ionic motion. However, the influence of such ionic dynamics on electronic transport properties has not been discussed explicitly. Consequently, the physical link between ionic dynamics and electronic scattering and transport processes during memristive switching remains unclear.

In this work, oxygen vacancies in TiO_2 are modeled as ionic particles carrying an effective charge of $+2q$. Based on an ion hopping model, we derive the time evolution equation for the state variable w . Furthermore, by numerically simulating the spatiotemporal distribution of oxygen vacancies formed by ionic motion, we model the scattering processes between electrons and oxygen vacancies and extract the average electronic scattering time τ . By applying the Drude model, we establish a quantitative relationship between the electronic scattering time τ and the resistance R , thereby reinterpreting the Strukov state variable w as a manifestation of a cascade of physical processes: ionic motion, modification of electronic scattering, and changes in transport properties. Ultimately, this work aims to propose a unified physical framework connecting the Strukov memristor model with Drude transport theory.

In many oxide-based memristors, such as TiO_2 devices, resistance modulation is commonly attributed to the motion of ionic defects, particularly oxygen vacancies, which alter the internal structure of the device under an applied electric field.

Conventional memristor models often describe the device state using a geometric variable $w(t)$ representing the extent of a doped region, with its evolution governed by phenomenological drift equations. While such descriptions successfully reproduce memristive current-voltage characteristics, the physical interpretation of the state variable remains largely structural, providing limited direct insight into how ionic motion modifies electronic transport.

In this work, we propose an alternative perspective that connects ionic dynamics to electronic transport through scattering physics. Building on this microscopic description, we introduce a transport-based interpretation in which the memristive state is characterized by the average of electronic scattering time, τ_w within the Drude framework.

By relating the scattering time to the transported charge through a closed-form expression, we demonstrate that the memristive state can be fully described by the charge history of the device. This formulation reinterprets the state variable not merely as a geometric quantity, but as an effective transport parameter, providing a physically transparent link between ionic motion, electron scattering, and resistance modulation.

2 Ion Hopping Model

2.1 dw/dt Equation

Assuming that ionic motion occurs via a thermally activated hopping process, the ionic mobility μ follows an Arrhenius-type relation

$$\mu \propto \nu_0 \exp\left(-\frac{E_\alpha}{kT}\right), \quad (2.1)$$

where E_α is the energy barrier between neighboring potential wells in the crystal lattice and ν_0 is the attempt frequency. Accordingly, the ionic hopping rate is defined as

$$\Gamma \equiv \nu_0 \exp\left(-\frac{E_\alpha}{kT}\right). \quad (2.2)$$

In the absence of an external electric field, the hopping probabilities in opposite directions are identical, and the system remains in a steady state. When a voltage V is applied across the memristor,

76 however, most of the potential drop occurs across the undoped region. The resulting external electric
77 field can therefore be approximated as

$$E_{\text{ext}}(w, V) = \frac{V}{D - w}. \quad (2.3)$$

78 For hopping along the lattice direction with lattice spacing d , the energy barriers in the presence of
79 an electric field become

$$E_{\alpha}^{\pm} = E_{\alpha 0} \mp qdE_{\text{ext}}(w), \quad (2.4)$$

80 leading to direction-dependent hopping rates

$$\Gamma_{\pm} = \nu_0 \exp\left(-\frac{E_{\alpha}^{\pm}}{kT}\right). \quad (2.5)$$

81 Since the state variable w represents the length of the doped region, its temporal evolution can be
82 written as

$$\frac{dw}{dt} = (\Gamma_+ - \Gamma_-)d. \quad (2.6)$$

83 An increase in w , however, corresponds to the accumulation of ionic defects and is generally
84 accompanied by an increase in the free energy of the system. To account for thermodynamic forces
85 opposing unlimited growth of the doped region, we incorporate contributions from both chemical
86 potential gradients [6] and elastic strain energy [7].

87 For an ionic concentration c , the chemical potential is given by

$$\mu_{\text{chem}} = \mu_0 + kT \ln c, \quad (2.7)$$

88 which implies that spatial variations in defect concentration generate an effective driving force on
89 the ions. By considering a small concentration fluctuation δc around a baseline c_0 , the gradient term
90 can be linearized as $\nabla \ln c \approx \nabla c/c_0$. Assuming the concentration gradient scales with the doped
91 region width w over total device length D (i.e., $\nabla c \approx \Delta c/D$), this effect is phenomenologically
92 represented as an effective electric field,

$$E_{\text{chem}} = -\frac{1}{q} \nabla \mu_{\text{chem}} \propto \frac{w}{D}. \quad (2.8)$$

93 In addition, the accumulation of oxygen vacancies induces local lattice distortion and elastic strain
94 ϵ . Within the Strukov framework, this strain is proportional to the ionic concentration, and thus
95 $\epsilon \propto w/D$. Within linear elasticity theory, the associated strain energy density scales quadratically
96 with strain:

$$U_{\text{strain}} = \frac{1}{2} C \epsilon^2 \propto \left(\frac{w}{D}\right)^2. \quad (2.9)$$

97 Accordingly, the effective electric fields associated with chemical potential gradients and elastic strain
98 accumulation are modeled as

$$E_{\text{chem}}(w) = A \frac{w}{D}, \quad E_{\text{strain}}(w) = B \left(\frac{w}{D}\right)^2. \quad (2.10)$$

99 where A and B are phenomenological parameters that quantify the strength of each contribution.

100 Combining these contributions, the time evolution of the state variable is finally obtained as

$$\frac{dw}{dt} = 2d\nu_0 \sinh\left(\frac{qdE_{\text{ext}}(w, V)}{kT}\right) \exp\left[-\frac{E_{\text{eff}}(w)}{kT}\right], \quad (2.11)$$

101 where $E_{\text{eff}}(w) = E_{\alpha 0} + E_{\text{chem}}(w) + E_{\text{strain}}(w)$ is the effective activation energy barrier incorporating
102 both chemical and elastic contributions. This formulation satisfies the definition of a Chua-type
103 memristive system (Equation (1.1)).

2.2 Modeling Framework and Dynamic Resistance Physics

The numerical simulation of the memristor is implemented using a Python-based solver that updates the device state and resistance at each time step using the Euler method. The device is modeled as two regions in series: a doped region of width $w(t)$ and an undoped region of width $D - w(t)$. The internal state evolution, representing the motion of the boundary between the two regions, is governed by the nonlinear ionic drift model introduced in (2.11).

In this framework, the resistances of the two regions are not treated as fixed material constants, but are dynamically determined by their respective dominant conduction mechanisms. This approach allows the macroscopic resistance modulation to be directly linked to microscopic transport parameters.

1. **Doped Region (R_{on}):** Conduction in the doped region is modeled using the Drude transport formalism, assuming a high concentration of oxygen vacancies that give rise to quasi-metallic behavior. The resistance of the doped region is expressed as

$$R_{\text{on}}(w) = \frac{w(t)m^*}{n_{\text{on}}q^2\tau_{\text{on}}A_{\text{area}}} \quad (2.12)$$

where m^* is the effective carrier mass, n_{on} is the carrier density in the doped region, τ_{on} is the defective momentum relaxation time and A_{area} is the device cross-sectional area.

The linear dependence of R_{on} on $w(t)$ arises purely from geometric scaling of the conductive path length, while the transport parameters are treated as effective quantities within the Drude approximation.

2. **Undoped Region (R_{off}):** The undoped region is assumed to be weakly conducting and dominated by thermally activated electron hopping. The hopping activation barrier is modeled as a state-dependent quantity,

$$E_{\alpha} = E_{\text{max}}(1 - \frac{w}{D}), \quad (2.13)$$

reflecting the progressive reduction of the insulating region as the doped boundary advances. This barrier determines the hopping rate and the corresponding mobility μ_{off} , yielding the resistance

$$R_{\text{off}}(w) = \frac{D - w(t)}{n_{\text{off}}q\mu_{\text{off}}A_{\text{area}}} \quad (2.14)$$

where n_{off} represents the lower carrier density of the undoped oxide. This formulation provides a minimal description of the transition from hopping-dominated to Drude-like transport as the internal state variable evolves.

The physical parameters employed in the simulation, summarized in Table 1, are chosen to be representative of TiO_2 -based memristive devices reported in the literature. In particular, the device thickness and intrinsic activation energy are consistent with nanoscale TiO_{2-x} films reported by [2] and [6]. The ionic attempt frequency ν_0 is selected to match typical lattice vibration frequencies in transition metal oxides [8]. The phenomenological barrier modulation parameters A and B , as well as the hopping distance (d), are chosen in accordance with field-enhanced transport models originally discussed by *Mott and Gurney*.

Table 1: Physical parameters and modeling constants for the memristor simulation.

Parameter	Value	Description / Reference
D	10 nm	Total thickness
kT	0.025 eV	Thermal energy at 300 K
ν_0	1×10^{13} Hz	Ionic attempt frequency
d	0.5 nm	Ion hopping distance
$E_{\alpha 0}$	0.7 eV	Intrinsic activation energy
A, B	0.2, 0.1 eV	Barrier modulation factors
$n_{\text{on}}, n_{\text{off}}$	$10^{27}, 10^{22} \text{ m}^{-3}$	Carrier concentrations
E_{max}	0.5 eV	Max electron hopping barrier

137 2.2.1 I-V Hysteresis Analysis

138 The simulated current-voltage (i-v) characteristics (Figure 1) demonstrate the characteristic pinched
 139 hysteresis loop. This behavior arises from the state-dependent conductance, where the device
 140 transitions between states as the ionic boundary w shifts under the external field.

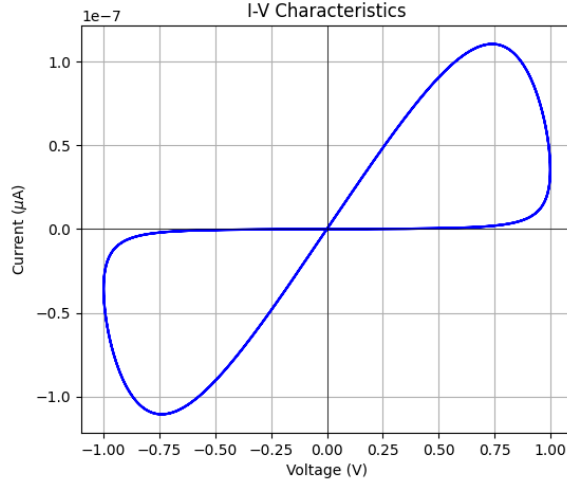


Figure 1: Simulated i-v characteristics showing the memristive hysteresis loop.

141 2.2.2 State Variable Evolution

142 Figure 2 illustrates the temporal evolution of the normalized state variable $w(t)/D$. Driven by the
 143 sinusoidal input, the doped region width oscillates, reflecting the physical redistribution of ions within
 144 the oxide layer. The nonlinear rate of change is a direct result of the exponential terms in the state
 145 equation (2.11).

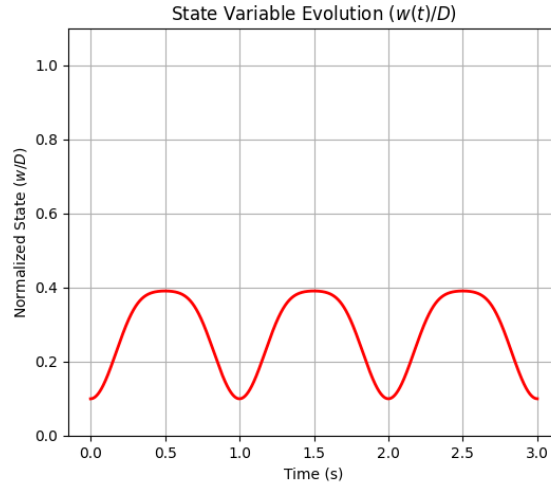


Figure 2: Evolution of the normalized state variable $w(t)/D$ over time.

2.2.3 Relaxation Time Dynamics

The relaxation dynamics shown in Figure 3 provide insight into the conduction speeds. While τ_{on} (Drude) remains constant at $\sim 10^{-12}$ s, the hopping-based τ_{off} varies dynamically between 10^{-9} and 10^{-7} s as a function of the internal energy barrier E_a .

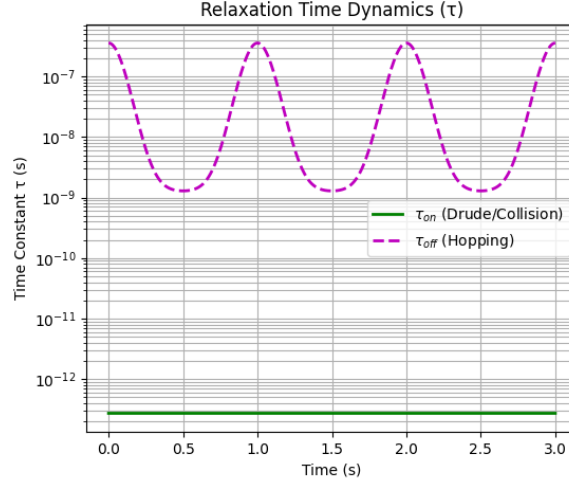


Figure 3: Comparison of relaxation times τ_{on} and τ_{off} on a log scale.

3 Electron Scattering

In previous sections, the ions in the TiO_2 memristors were modeled as hopping ionic particles carrying a charge, and the evolution of the state variable $w(t)$ was derived based on the dynamics of the thermally activated ion hopping dynamics. By incorporating not only the external electric field but also contributions from chemical potential gradients and elastic strain energy, we demonstrated that ionic motion leads to a nonlinear dynamics of the structural state variable. Numerical simulations further confirmed the resistance modulation induced by ionic redistribution, reproducing the characteristic memristive hysteresis behavior.

However, this description of ionic dynamics remains indirect from the perspective of electronic transport. In practice, electrical current is not carried by the ions themselves, but by electrons moving through a potential landscape shaped by the ionic distribution. Therefore, a more physical understanding of memristive resistance modulation requires an explicit connection between the outcomes of ionic motion and the resulting electronic transport properties.

In this section, we reinterpret the results of the ion hopping dynamics derived above in the language of electronic transport, and discuss how structural changes in the doped region can be summarized in terms of electron scattering characteristics, in particular the average scattering time τ with two key postulates

3.1 Postulates for τ -Based Modeling

In this subsection, we introduce two phenomenological postulates that allow the results of ion hopping dynamics to be translated into electronic transport properties. Rather than explicitly modeling microscopic electron-ion interactions, we adopt a coarse-grained description in which the influence of ionic redistribution is summarized by the average scattering time τ . These postulates provide the minimal assumptions required to relate the variable of the structural state $w(t)$ to electronic scattering and resistance.

3.1.1 Postulate I, Scattering centers Increase with the Doped Region Length

We assume that the total number of scattering centers N_s increases proportionally with the length of the doped region $w(t)$. Here, a scattering center denotes any dopant-related defect or ionic configuration that contributes to electron momentum relaxation. Under the assumption of an approximately uniform density of such scattering sources, the total number of scattering centers scales linearly with the spatial extent of the doped region.

Then, this proportional relation is expressed as

$$N_s = N_0 + \beta w(t) \quad (3.1)$$

(where β is a constant.)

At time $t = 0$, let $w(t = 0)$ and $N_s = N_0$ be equal to 0). Accordingly, the relation can be finally written as

$$N_s = \beta w(t) \quad (3.2)$$

3.1.2 Postulate II, Scattering Rate Scales with the number of scatterers

We further assume that the total electron scattering rate Γ_w is proportional to the number of scattering centers N_s . This assumption follows the Drude picture of independent scattering events, in which individual scattering contributions add linearly. therefore, the average scattering time τ_w decreases as the number of scattering centers increases. ($\Gamma_w \propto N_s$)

With (3.2), this relation is expressed as

$$\Gamma_w = \Gamma_0 + \alpha w \quad (3.3)$$

Since the average scattering time τ_w is the inverse of the scattering rate Γ_w , we can re-express τ_w in terms of Γ_0 and w as follows.

$$\tau_w(w) = \frac{1}{\Gamma_0 + \alpha w} \quad (3.4)$$

3.2 Expressing $w(t)$ by small E_{ext} field approximation

3.2.1 Approximation A: Small External E-Field, E_{ext}

we have $E_{\text{ext}} \sinh \frac{qdE_{\text{ext}}}{kT}$, in (2.11)

In case there is no voltage exerted ($|\frac{qd}{kT} E_{\text{ext}}| \ll 1$, Then $\sinh(\frac{qdE_{\text{ext}}(w,V)}{kT})$ is approximately expressed as $\frac{qd}{kT} E_{\text{ext}}$

3.2.2 Approximation B: Small fraction $\frac{w(t)}{D}$

In (2.11), we have $E_{\text{chem}}(w)$ and $E_{\text{strain}}(w)$ which depend on $w(t)$. If we consider a small change in $w(t)$, which means $\frac{w}{D} \ll 1$ and $D - w \approx D$, It can be approximated as a constant for (2.11) since there's only dependence on $w(t)$.

Accordingly, we get

$$\frac{dw}{dt} \approx \mu_{\text{ion}} E_{\text{ext}} \quad (3.5)$$

where μ_{ion} denotes $2d^2\nu_0 \frac{q}{kT} \exp[-\frac{E_{\alpha 0} + E_{\text{chem}} + E_{\text{strain}}}{kT}]$.

Since $E_{\text{ext}}(w, V) = \frac{V}{D-w}$, we can express (3.5) with approximation B as follows

$$\frac{dw}{dt} \approx \mu_{\text{ion}} E_{\text{ext}} = \mu_{\text{ion}} \frac{V}{D} = \mu_{\text{ion}} \frac{R}{D} I(t). \quad (3.6)$$

According to [2], $w(t) = \mu_V \frac{R_{\text{on}}}{D}$ where $\mu_V \frac{R_{\text{on}}}{D} \approx \mu_{\text{ion}} \frac{R}{D}$

205 Then we finally obtain

$$w(t) \approx \mu_V \frac{R_{\text{on}}}{D} q(t) \quad (3.7)$$

206 3.3 τ_w as a state variable closed by q

207 To obtain an explicit expression of the scattering time in terms of the transported charge q , we
 208 consider the total derivative of the average scattering time with respect to q . This approach allows
 209 the structural and transport dynamics, previously expressed through the state variable $w(t)$, to be
 210 reformulated in a closed form governed solely by the charge history.

211 Closing the description in terms of q is essential, as a memristive system is fundamentally defined
 212 by a constitutive relation in which the state of devices is uniquely determined by the charge that has
 213 flowed through it. By expressing the average scattering time τ_w as a function of q , we reinterpret
 214 the memristive state not as a purely geometric variable, but as an effective transport parameter that
 215 directly encodes the history-dependent electronic response of the device in classical way.

216 First, we expressed $\frac{d\tau_w}{dq}$ with the chain rule as

$$\frac{d\tau_w}{dq} = \frac{d\tau_w}{dw} \frac{dw}{dq} \quad (3.8)$$

217 If we take a total derivative, $\frac{d\tau_w}{dw}$ from (3.4), we get

$$\frac{d\tau_w}{dw} = \frac{d}{dw} \left(\frac{1}{\Gamma_0 + \alpha w} \right) = -\alpha (\Gamma_0 + \alpha w)^{-2} \quad (3.9)$$

218 which is equal to

$$\frac{d\tau_w}{dw} = -\alpha \tau_w^2 \quad (3.10)$$

219 And if we take a total derivative of $\frac{dw}{dq}$, with (3.7)

220 we get

$$\frac{dw}{dq} = \mu_V \frac{R_{\text{on}}}{D} \quad (3.11)$$

221 Therefore, (3.8) can be expressed as

$$\frac{d\tau_w}{dq} = -\alpha \tau_w^2 \mu_V \frac{R_{\text{on}}}{D} = -\frac{\alpha \mu_V R_{\text{on}}}{D} (\Gamma_0 + \frac{\alpha \mu_V R_{\text{on}}}{D} q)^{-2} \quad (3.12)$$

222 Then if we take integral on each sides of (3.12),

$$\tau_w = -\frac{\alpha \mu_V R_{\text{on}}}{D} \int (\Gamma_0 + \frac{\alpha \mu_V R_{\text{on}}}{D} q)^{-2} dq \quad (3.13)$$

223 which is calculated with proper initial condition, as follows

$$\tau_w(q) = -\frac{\alpha \mu_V R_{\text{on}}}{D} \left(\frac{\alpha \mu_V R_{\text{on}}}{D} \right)^{-1} (\Gamma_0 + \frac{\alpha \mu_V R_{\text{on}}}{D} q)^{-1} \quad (3.14)$$

224 Then we get

$$\tau_w(q) = (\Gamma_0 + \frac{\alpha \mu_V R_{\text{on}}}{D} q)^{-1} \quad (3.15)$$

225 Therefore, the average scattering time τ_w is closed with respect to the charge q , allowing it to serve
 226 as a state variable governed solely by the charge history of the device and also to show axiomatically
 227 memristive behavior.

4 Conclusion

In this work, we presented a unified physical framework that connects ionic motion and electronic transport in TiO_2 -based memristors. Oxygen vacancies were modeled as charged ionic defects, and their thermally activated hopping dynamics were used to derive a nonlinear evolution equation for the structural state variable $w(t)$. By incorporating the effects of external electric fields, chemical potential gradients, and elastic strain energy, the model captures the essential physics governing ionic redistribution and reproduces characteristic memristive hysteresis through numerical simulation.

To establish an explicit link between ionic dynamics and electronic transport, we reinterpreted the structural state variable in terms of electron scattering. Introducing two minimal phenomenological postulates, we related the doped region length to the number of scattering centers and, consequently, to the electron scattering rate within the Drude transport picture. This allowed the macroscopic resistance to be expressed through the average electronic scattering time τ , providing a transport-based interpretation of the memristive state.

Furthermore, by adopting the constitutive relation of the Strukov model for the state evolution and applying the total derivative with respect to the transported charge q , we demonstrated that the average scattering time τ_w can be expressed as a closed-form function of q . This result shows that the memristive state can be fully characterized by the charge history of the device, satisfying the defining condition of a memristive system.

Overall, this study shifts the interpretation of the memristive state variable from a purely geometric quantity to an effective transport parameter rooted in microscopic scattering physics. The proposed framework provides a physically transparent bridge between ionic motion, electron scattering, and macroscopic resistance modulation, and offers a foundation for extending memristor modeling toward more transport-oriented and material-specific descriptions.

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A Appendix / supplemental material

A.1 Simulation Code

The complete Python implementation used to generate the results in this study is provided below. The code calculates the state evolution based on Equation (2.11) and derives the resistance using the Drude and Hopping models.

```
import numpy as np
import matplotlib.pyplot as plt

# -----
# 1. Physical Constants and Parameters (SI Units)
# -----
q = 1.6e-19          # Elementary charge (C)
m_eff = 9.1e-31      # Effective electron mass (kg)
k_B = 1.38e-23       # Boltzmann constant (J/K)
eV_to_J = 1.6e-19

T = 300              # Temperature (K)
D = 10e-9             # Device length (10 nm)
A_area = 100e-18      # Area (100 nm^2)

# Input: 1.0V Sine wave at 1.0Hz
time = np.arange(0, 3.0, 1e-5)
V_signal = 1.0 * np.sin(2 * np.pi * 1.0 * time)

# Ion Dynamics (Ref: Equation 2.11)
d, nu_ion = 0.5e-9, 1e13
E_alpha0 = 0.7 * eV_to_J
A_bar, B_bar = 0.2 * eV_to_J, 0.1 * eV_to_J

# Carrier Densities
n_on, n_off = 1e27, 1e22
```

```

303 # Physics Engine Constants
304 a_radius = 0.1e-9
305 v_th = np.sqrt(3 * k_B * T / m_eff)
306 E_max = 0.5 * eV_to_J
307 nu_elec, d_elec = 1e14, 0.5e-9
308
309 # -----
310 # 2. Initialization and Main Simulation Loop (Euler Method)
311 # -----
312 N = len(time)
313 w, current = np.zeros(N), np.zeros(N)
314 tau_on_trace, tau_off_trace = np.zeros(N), np.zeros(N)
315 w[0] = 1.0e-9
316
317 for i in range(N - 1):
318     # A. State Update (Equation 2.11)
319     gap = max(D - w[i], 1e-12)
320     E_ext = V_signal[i] / gap
321     w_norm = w[i] / D
322     E_barrier = E_alpha0 + A_bar * w_norm + B_bar * (w_norm**2)
323
324     dw_dt = 2 * d * nu_ion * np.exp(-E_barrier/(k_B*T)) * np.sinh(q*d*E_ext/(k_B*T))
325     w[i+1] = np.clip(w[i] + dw_dt * 1e-5, 1e-12, D - 1e-12)
326
327     # B. Resistance and tau calculation
328     # 1. Doped Region (Drude)
329     tau_on = 1 / (n_on * (np.pi * a_radius**2) * v_th)
330     R_on = (w[i] * m_eff) / (n_on * (q**2) * tau_on * A_area)
331
332     # 2. Undoped Region (Hopping)
333     E_a = E_max * (1 - w[i]/D)
334     tau_off = 1.0 / (nu_elec * np.exp(-E_a / (k_B * T)))
335     mu_off = (q * d_elec**2) / (k_B * T * tau_off)
336     R_off = (D - w[i]) / (n_off * q * mu_off * A_area)
337
338     current[i] = V_signal[i] / (R_on + R_off)
339     tau_on_trace[i], tau_off_trace[i] = tau_on, tau_off
340
341 # -----
342 # 4. Visualization (Plotting)
343 # -----
344 # Figure 1: I-V Curve
345 plt.figure(figsize=(6, 5))
346 plt.plot(V_signal, current * 1e6, 'b-')
347 plt.title('I-V Characteristics')
348 plt.xlabel('Voltage (V)'); plt.ylabel('Current (uA)'); plt.grid(True); plt.show()
349
350 # Figure 2: State evolution
351 plt.figure(figsize=(6, 5))
352 plt.plot(time, w / D, 'r-')
353 plt.title('State Variable Evolution ($w(t)/D$)')
354 plt.xlabel('Time (s)'); plt.ylabel('Normalized State ($w/D$)'); plt.grid(True); plt.show()
355
356 # Figure 3: tau comparison
357 plt.figure(figsize=(6, 5))
358 plt.semilogy(time, tau_on_trace, 'g-', label='tau_on (Drude)')
359 plt.semilogy(time, tau_off_trace, 'm--', label='tau_off (Hopping)')
360 plt.title('Relaxation Time Dynamics (tau)')
361 plt.xlabel('Time (s)'); plt.ylabel('Time Constant tau (s)'); plt.legend(); plt.grid(True); plt.show()

```

AI Co-Scientist Challenge Korea Paper Checklist

1. Claims

Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?

Answer: [Yes]

Justification: The paper reports numerical simulation results based on the physical mechanisms of TiO_2 memristors, which are consistent with the provided physics engine code and resulting graphs.

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